

HU Collection

TMUA Mock Set E Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	B	A	D	E	B	F	C	D	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	C	E	F	D	E	B	F	A	D

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer C.

Work with the symmetric quantities rather than solving. $x^3 + y^3 = (x + y)^3 - 3xy(x + y)$ gives $40 = 64 - 12xy$, so $xy = 2$. Then $x^2 + y^2 = 16 - 4 = 12$, and $x^4 + y^4 = (x^2 + y^2)^2 - 2x^2y^2 = 144 - 8 = 136$. (The numbers are $2 \pm \sqrt{2}$, which are real, so the data are consistent.)

Question 2. Answer B.

With $t = \log_2 x$, we have $\log_x 4 = \frac{2}{t}$, so $\frac{4}{t} - t = 1$, that is $t^2 + t - 4 = 0$. Both roots are real and non-zero, giving two valid x (positive, and not equal to 1). The product of the solutions is $2^{t_1} \cdot 2^{t_2} = 2^{t_1+t_2} = 2^{-1} = \frac{1}{2}$ by Vieta, without finding either root.

Question 3. Answer A.

The perpendicular through $(1, 2)$ has gradient $-\frac{1}{2}$: $y = -\frac{1}{2}x + \frac{5}{2}$. It meets $y = 2x - 3$ where $\frac{5}{2}x = \frac{11}{2}$, at $(\frac{11}{5}, \frac{7}{5})$, the midpoint of the point and its image. So the image is $(2 \cdot \frac{11}{5} - 1, 2 \cdot \frac{7}{5} - 2) = (\frac{17}{5}, \frac{4}{5})$. Option C is the foot of the perpendicular, a common place to stop.

Question 4. Answer D.

Rationalise: $\frac{1}{\sqrt{k} + \sqrt{k+2}} = \frac{\sqrt{k+2} - \sqrt{k}}{2}$. The sum telescopes with a gap of two, so two terms survive at each end:

$$\frac{1}{2} \left(\sqrt{50} + \sqrt{49} - \sqrt{1} - \sqrt{2} \right) = \frac{1}{2} \left(5\sqrt{2} + 7 - 1 - \sqrt{2} \right) = 3 + 2\sqrt{2}.$$

Keeping only one surviving term at each end, as for a gap of one, gives the wrong answer.

Question 5. Answer E.

Let $c_k = \binom{10}{k} 2^{10-k} 3^k$. Then $\frac{c_{k+1}}{c_k} = \frac{10-k}{k+1} \cdot \frac{3}{2}$, which is at least 1 exactly when $3(10-k) \geq 2(k+1)$, that is $k \leq 5.6$. So the coefficients increase up to c_6 and decrease after it: the greatest is at $k = 6$. The middle of the expansion, $k = 5$, is where $\binom{10}{k}$ peaks, but the factor $\frac{3}{2}$ pushes the peak one place right.

Question 6. Answer B.

Let $g(x) = 2^x + x^2 - 3$. Then $g(1) = 0$ exactly, $g(-1) = -\frac{3}{2} < 0$ and $g(-2) = \frac{5}{4} > 0$, so there is a second root in $(-2, -1)$. There are no more, because g is the sum of 2^x and x^2 , both of which curve upwards, so the graph of g is a single valley and crosses any horizontal line at most twice. Exactly two solutions. The root $x = 1$ is easy to spot; the negative one is the one people miss.

Question 7. Answer F.

$\frac{a}{1-r} = 12$ and $\frac{a^2}{1-r^2} = 48$. Divide the second by the first: $\frac{a}{1+r} = 4$. So $a = 12(1-r) = 4(1+r)$, giving $r = \frac{1}{2}$ and $a = 6$. The cubes form a geometric series with first term 216 and ratio $\frac{1}{8}$, with sum $\frac{216}{7/8} = \frac{1728}{7}$. Option E comes from using $1 - r^3 = \frac{5}{8}$.

Question 8. Answer C.

Squaring gives $1 + 2\sin x \cos x = \frac{49}{25}$, so $\sin x \cos x = \frac{12}{25}$. Then $\sin x$ and $\cos x$ are the roots of $t^2 - \frac{7}{5}t + \frac{12}{25} = 0$, namely $\frac{3}{5}$ and $\frac{4}{5}$, in either order. Both orders give angles in the first quadrant, so $\tan x = \frac{3}{4}$ or $\frac{4}{3}$, and the sum is $\frac{25}{12}$. The trap is to find one value and stop.

Question 9. Answer D.

The curves meet where $x^3 = 4x$, at $x = -2, 0, 2$. By the odd symmetry of both curves the two regions have equal area, and the one for $0 \leq x \leq 2$ is $\int_0^2 (4x - x^3) dx = 8 - 4 = 4$. The total is 8. Integrating $4x - x^3$ from -2 to 2 gives 0, the signed cancellation that the word “total” warns against.

Question 10. Answer B.

Completing the square, $C = (1, 2)$ and the radius is 3. $CP^2 = 36 + 1 = 37$, so each tangent has length $\sqrt{37-9} = 2\sqrt{7}$. The quadrilateral consists of two right-angled triangles CAP and CBP , each of area $\frac{1}{2} \cdot 3 \cdot 2\sqrt{7}$, so the total is $6\sqrt{7}$.

Question 11. Answer A.

If $k = 1$ the expression is $2x + 4$, which is negative for $x < -2$, so $k = 1$ fails. Otherwise we need a positive leading coefficient, $k > 1$, and a non-positive discriminant: $4k^2 - 4(k-1)(k+3) = 12 - 8k \leq 0$, that is $k \geq \frac{3}{2}$. So $k \geq \frac{3}{2}$, and the boundary is included, since a zero discriminant still satisfies ≥ 0 .

Question 12. Answer C.

Eliminate $y = \frac{32}{x^2}$ and minimise $g(x) = x + \frac{64}{x^2}$ for $x > 0$. $g'(x) = 1 - \frac{128}{x^3} = 0$ at $x^3 = 128$, so $x = 4\sqrt[3]{2}$, and this is a minimum since $g \rightarrow \infty$ at both ends. Then $\frac{64}{x^2} = \frac{64}{16\sqrt[3]{4}} = \frac{4}{\sqrt[3]{4}} = 2\sqrt[3]{2}$, so the minimum is $4\sqrt[3]{2} + 2\sqrt[3]{2} = 6\sqrt[3]{2}$. At the minimum $x = 4y$, the same balance a three-term AM-GM with $\frac{x}{2} + \frac{x}{2} + 2y$ would predict.

Question 13. Answer E.

With $s = \sin x \in [-1, 1]$, the equation is $k = 1 - s^2 - s = \frac{5}{4} - (s + \frac{1}{2})^2$. On $[-1, 1]$ this takes its maximum $\frac{5}{4}$ at $s = -\frac{1}{2}$ and its minimum -1 at $s = 1$, and every value in between. So $-1 \leq k \leq \frac{5}{4}$. Allowing all real s gives F; forgetting the vertex gives A.

Question 14. Answer F.

The area is $\frac{1}{2}|4 \cdot 5 - 1 \cdot k| = \frac{1}{2}|20 - k|$. Setting it equal to 10 gives $20 - k = \pm 20$, so $k = 0$ or $k = 40$, and the sum is 40. For $k = 0$ the third vertex is $(0, 5)$, with base 5 on the y -axis and height 4: area 10, as required. Missing the modulus finds only one value.

Question 15. Answer D.

The two centres and either intersection point form an equilateral triangle, so each centre sees the common chord at an angle of $\frac{2\pi}{3}$. The lens is two circular segments, each a sector of area $\frac{1}{2} \cdot \frac{2\pi}{3} = \frac{\pi}{3}$ minus a triangle of area $\frac{1}{2} \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{4}$. The total is $2 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$. Option A is one segment only.

Question 16. Answer E.

The tangent gradient at P is 2, so the normal is $y - 1 = -\frac{1}{2}(x - 1)$. Substituting into $y = x^2$ gives $2x^2 + x - 3 = 0$, that is $(x - 1)(2x + 3) = 0$, so $Q = (-\frac{3}{2}, \frac{9}{4})$. Then $PQ = \sqrt{(\frac{5}{2})^2 + (\frac{5}{4})^2} = \frac{5}{4}\sqrt{4 + 1} = \frac{5\sqrt{5}}{4}$.

Question 17. Answer B.

The integral is the signed area under $y = x^2 - 4x$ over a window of width 1, and it is smallest when the window sits where the curve is lowest. The curve is symmetric about its minimum at $x = 2$, so the best window is $[\frac{3}{2}, \frac{5}{2}]$, giving $a = \frac{3}{2}$. To confirm, $I(a) = \frac{(a+1)^3 - a^3}{3} - 2((a+1)^2 - a^2) = a^2 - 3a - \frac{5}{3}$, which is minimised at $a = \frac{3}{2}$. Choosing $a = 2$ puts the minimum at the edge of the window rather than its centre.

Question 18. Answer F.

$f'(x) = 3(x^2 - a^2)$, so f decreases on $[0, a]$ and increases after a . On $[0, 2]$ the maximum is therefore at an endpoint: it is $\max(f(0), f(2)) = \max(0, 8 - 6a^2)$. This equals 0 exactly when $8 - 6a^2 \leq 0$, that is $a \geq \frac{2}{\sqrt{3}}$. It holds even when $a > 2$, since then f decreases on the whole interval.

Question 19. Answer A.

The curve is below the axis on $[1, 2)$ and above it on $(2, 4]$. With $F(x) = \frac{2}{3}x^{3/2} - 4x^{1/2}$: $F(1) = -\frac{10}{3}$, $F(2) = -\frac{8\sqrt{2}}{3}$ and $F(4) = -\frac{8}{3}$. The two areas are $F(1) - F(2) = \frac{8\sqrt{2}-10}{3}$ and $F(4) - F(2) = \frac{8\sqrt{2}-8}{3}$, totalling $\frac{16\sqrt{2}-18}{3}$. The signed integral from 1 to 4 is $\frac{2}{3}$, option B.

Question 20. Answer D.

Complete the square: $u_{n+1} - 1 = (u_n - 1)^2$. So with $v_n = u_n - 1$ the recurrence is $v_{n+1} = v_n^2$, giving $v_n = v_1^{2^{n-1}}$. This stays bounded exactly when $|v_1| \leq 1$, that is $0 \leq u_1 \leq 2$: then $|v_n| \leq 1$ throughout. The endpoints behave differently, since $u_1 = 0$ and $u_1 = 2$ both lead to the constant sequence 2, but both are bounded. For $|v_1| > 1$ the terms grow without limit.